

Entrega 2 Semana 5: Implementacion de Jenkins en SkillCert

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Enfasis Profesional I - Integracion Continua

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Resumen

Esta entrega implementa Jenkins como gestor de operaciones de integracion continua para SkillCert. El pipeline valida el repositorio con checkout desde GitHub, instalacion de dependencias, pruebas backend, build frontend, validacion de Docker Compose y construccion de contenedores.

Objetivo

Aplicar integracion continua mediante Jenkins y demostrar con evidencias visuales que el pipeline se ejecuto correctamente.

Repositorio

<https://github.com/Jonevan369/skillcert-integracion-continua>

Caracteristicas requeridas para Jenkins

Caracteristica	Implementacion
Jenkins	Jenkins LTS 2.541.3 ejecutado en contenedor temporal.
Node.js	NodeJS-20 configurado en Jenkins; consola evidencia v20.20.2.
Docker	Docker Desktop activo y socket Docker montado en el agente Jenkins.
Build tools	build-essential instalado para compilar dependencias nativas como better-sqlite3.
Plugins	Pipeline, Git, GitHub Branch Source, Pipeline Stage View, NodeJS, Docker Workflow y Timestampper.
Repositorio	GitHub: https://github.com/Jonevan369/skillcert-integracion-continua

Pipeline ejecutado

Checkout, Install Dependencies, Backend Tests, Frontend Build, Validate Docker Compose, Build Containers y Post Actions terminaron en SUCCESS.

Evidencias visuales

Captura 1. Dashboard Jenkins con job skillcert-ci y ultimo exito

+ Nueva Tarea

Historial de trabajos

Trabajos en la cola

No hay trabajos en la cola

Estado del ejecutor de construcciones

0/2

Todo

+

S	W	Nombre ↓	Último Éxito	Último Fallo	Última Duración
		skillcert-ci	1 Min 57 Seg #1	N/D	1 Min 0 Seg

Icono:

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🔔

Sign in

añadir descripción

REST API Jenkins 2.541.3

Captura 2. Build Jenkins exitoso con repositorio y duracion

skillcert-ci

#1

🔍

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Sign in

añadir descripción

Conservar esta ejecución para siempre

Status

</> Changes

Console Output

Edit Build Information

Delete build '#1'

Git Build Data

Restart from Stage

Replay

Pipeline Steps

Workspaces

#1 (6 jun 2026 23:13:50)

git

Revision: 7c471d2484a411e04861eef2894ae8f71e8c6207

Repository: <https://github.com/Jonevan369/skillcert-integracion-continua>

• refs/remotes/origin/main

</>

Sin cambios

Comenzó hace 1 Min 58 Seg

Tardó 1 Min 0 Seg

REST API Jenkins 2.541.3

Captura 3. Etapas del pipeline en SUCCESS desde Jenkins wfapi

Jenkins Pipeline - skillcert-ci #1 - SUCCESS

Duracion total: 60.0 s | Fuente: <http://localhost:8080/job/skillcert-ci/1/wfapi/describe>

Etapas	Estado	Duracion
Declarative: Tool Install	 SUCCESS	0.07 s
Checkout	 SUCCESS	4.10 s
Install Dependencies	 SUCCESS	48.38 s
Backend Tests	 SUCCESS	0.34 s
Frontend Build	 SUCCESS	1.92 s
Validate Docker Compose	 SUCCESS	0.32 s
Build Containers	 SUCCESS	3.15 s
Declarative: Post Actions	 SUCCESS	0.29 s

Captura 4. Consola Jenkins: npm test -w server y pass 3

Consola Jenkins - npm test -w server y 3 pruebas aprobadas

```
[Pipeline] stage
[Pipeline] { (Backend Tests)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
[2026-06-06T23:14:44.324Z] + npm test -w server
[2026-06-06T23:14:44.324Z]
[2026-06-06T23:14:44.324Z] > @competence-platform/server@0.1.0 test
[2026-06-06T23:14:44.325Z] > node --test src/tests/*.test.js
[2026-06-06T23:14:44.325Z]
[2026-06-06T23:14:44.325Z] TAP version 13
[2026-06-06T23:14:44.325Z] # Subtest: eligibleBadgesForSkill assigns badges at configured thresholds
[2026-06-06T23:14:44.325Z] ok 1 - eligibleBadgesForSkill assigns badges at configured thresholds
[2026-06-06T23:14:44.325Z] ---
[2026-06-06T23:14:44.325Z] duration_ms: 0.878375
[2026-06-06T23:14:44.325Z] ...
[2026-06-06T23:14:44.325Z] # Subtest: voteWeightForKarma caps reputation influence
[2026-06-06T23:14:44.325Z] ok 2 - voteWeightForKarma caps reputation influence
[2026-06-06T23:14:44.325Z] ---
[2026-06-06T23:14:44.325Z] duration_ms: 1.658542
[2026-06-06T23:14:44.325Z] ...
[2026-06-06T23:14:44.325Z] # Subtest: calculateKarmaDelta rewards consensus alignment and penalizes disagreement lightly
[2026-06-06T23:14:44.325Z] ok 3 - calculateKarmaDelta rewards consensus alignment and penalizes disagreement lightly
[2026-06-06T23:14:44.325Z] ---
[2026-06-06T23:14:44.325Z] duration_ms: 0.11425
[2026-06-06T23:14:44.325Z] ...
[2026-06-06T23:14:44.325Z] 1..3
[2026-06-06T23:14:44.325Z] # tests 3
[2026-06-06T23:14:44.325Z] # suites 0
[2026-06-06T23:14:44.325Z] # pass 3
[2026-06-06T23:14:44.325Z] # fail 0
[2026-06-06T23:14:44.325Z] # cancelled 0
[2026-06-06T23:14:44.325Z] # skipped 0
[2026-06-06T23:14:44.325Z] # todo 0
[2026-06-06T23:14:44.325Z] # duration_ms 44.855458
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Frontend Build)
[Pipeline] tool
[Pipeline] envVarsForTool
```

Captura 5. Consola Jenkins: docker ps, docker compose build e imagenes Built

Consola Jenkins - Docker habilitado y contenedores contruidos

```
[2026-06-06T23:13:56.188Z] + docker ps
[2026-06-06T23:13:56.189Z] CONTAINER ID        IMAGE                                     COMMAND                                CREATED      STATUS      PORTS
[2026-06-06T23:13:56.189Z] 1928b89645ff    skillcert-jenkins:local                "/usr/bin/tini -- /u..."           17 seconds ago    Up 17 seconds    0.0.0.0:8080
[2026-06-06T23:14:46.886Z] + docker compose build
[2026-06-06T23:14:49.735Z] Image skillcert_ci_1-server Built
[2026-06-06T23:14:49.735Z] Image skillcert_ci_1-client Built
[2026-06-06T23:14:50.081Z] SkillCert CI pipeline completed successfully.
Finished: SUCCESS
```

Captura 6. Jenkinsfile publicado en GitHub

The screenshot shows the GitHub interface for the repository 'skillcert-integracion-continua' by user 'Jonesvan369'. The 'Jenkinsfile' is selected in the file browser on the left. The main area displays the Jenkinsfile code, which is a Jenkins pipeline script. The code defines a pipeline with an agent 'any', tools for NodeJS, and options for timestamps, disabling concurrent builds, and build discarding. The environment section sets 'NODE_ENV' to 'test' and 'COMPOSE_PROJECT_NAME' to 'skillcert_ci_\${env.BUILD_NUMBER}'. The pipeline consists of two stages: 'Checkout' which checks out the 'main' branch from the repository, and 'Install Dependencies' which runs 'npm ci'.

```
1 pipeline {
2   agent any
3
4   tools {
5     nodejs 'NodeJS-20'
6   }
7
8   options {
9     timestamps()
10    disableConcurrentBuilds()
11    buildDiscarder(logRotator(numToKeepStr: '10'))
12  }
13
14  environment {
15    NODE_ENV = 'test'
16    COMPOSE_PROJECT_NAME = "skillcert_ci_${env.BUILD_NUMBER}"
17  }
18
19  stages {
20    stage('Checkout') {
21      steps {
22        git branch: 'main',
23          url: 'https://github.com/Jonesvan369/skillcert-integracion-continua'
24      }
25    }
26
27    stage('Install Dependencies') {
28      steps {
29        sh 'node --version && npm --version'
30        sh 'npm ci'
31      }
32    }
29
```

Solucion del error de Docker

El error de permisos al Docker local se resolvió dando al agente Jenkins acceso al daemon Docker. En esta evidencia se usó Jenkins en contenedor con el socket Docker montado y usuario root.

```
docker run -u root -v /Users/johnvanegas/.docker/run/docker.sock:/var/run/docker.sock skillcert-jenkins:local
```

Equivalente Linux: `sudo usermod -aG docker jenkins && sudo systemctl restart jenkins`

Webhook GitHub

La URL requerida para GitHub es `http://SERVIDOR_JENKINS/github-webhook/`. La captura real del webhook no se pudo completar desde este entorno porque GitHub solicitó sesión/permisos de administrador al abrir Settings > Webhooks. No se incluye una captura falsa; debe completarse desde la cuenta propietaria del repositorio.

Anexo 1: Jenkinsfile completo

```
pipeline {
  agent any

  tools {
```

```

    nodejs 'NodeJS-20'
  }

  options {
    timestamps()
    disableConcurrentBuilds()
    buildDiscarder(logRotator(numToKeepStr: '10'))
  }

  environment {
    NODE_ENV = 'test'
    COMPOSE_PROJECT_NAME = "skillcert_ci_${env.BUILD_NUMBER}"
  }

  stages {
    stage('Checkout') {
      steps {
        git branch: 'main',
            url: 'https://github.com/Jonevan369/skillcert-integracion-continua'
      }
    }

    stage('Install Dependencies') {
      steps {
        sh 'node --version && npm --version'
        sh 'npm ci'
      }
    }

    stage('Backend Tests') {
      steps {
        sh 'npm test -w server'
      }
    }

    stage('Frontend Build') {
      steps {
        sh 'npm run build'
      }
    }

    stage('Validate Docker Compose') {
      steps {
        sh 'docker ps'
        sh 'docker compose config'
      }
    }

    stage('Build containers') {
      steps {
        sh 'docker compose build'
      }
    }
  }

  post {
    always {
      sh 'docker compose down --volumes --remove-orphans || true'
    }
    success {
      echo 'SkillCert CI pipeline completed successfully.'
    }
    failure {
      echo 'SkillCert CI pipeline failed. Review the failing stage logs in Jenkins.'
    }
  }
}

```

Anexo 2: Extracto de consola Jenkins

```

[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
[2026-06-06T23:14:44.324Z] + npm test -w server
[2026-06-06T23:14:44.324Z]
[2026-06-06T23:14:44.324Z] > @competence-platform/server@0.1.0 test
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[2026-06-06T23:14:44.325Z]   duration_ms: 1.658542
[2026-06-06T23:14:44.325Z]   ...
[2026-06-06T23:14:44.325Z] # Subtest: calculateKarmaDelta rewards consensus alignment and penalizes disagreement lightly

```

```
[2026-06-06T23:14:44.325Z] ok 3 - calculateKarmaDelta rewards consensus alignment and penalizes disagreement lightly
[2026-06-06T23:14:44.325Z] ---
[2026-06-06T23:14:44.325Z]   duration_ms: 0.11425
[2026-06-06T23:14:44.325Z]   ...
[2026-06-06T23:14:44.325Z] 1..3
[2026-06-06T23:14:44.325Z] # tests 3
[2026-06-06T23:14:44.325Z] # suites 0
[2026-06-06T23:14:44.325Z] # pass 3
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[2026-06-06T23:14:44.325Z] # cancelled 0
[2026-06-06T23:14:44.325Z] # skipped 0
[2026-06-06T23:14:44.325Z] # todo 0
[2026-06-06T23:14:44.325Z] # duration_ms 44.855458
[Pipeline] }
```

Conclusiones

La integracion continua quedo aplicada y demostrada con Jenkins: el job clono el repositorio, instalo dependencias, ejecuto pruebas, compilo el frontend, valido Docker Compose y construyo los contenedores. La ejecucion final termino en SUCCESS y quedo soportada con capturas y consola exportada.